



OFFICIAL

Acute Coronary Occlusion Myocardial Infarction (ACOMI) and Reperfusion Strategy Guideline

Table of Contents

1. Purpose	2
2. Applicability	2
3. Guidelines	2
3.1 Vital signs	2
3.2 ECG	2
3.2.1 Initial ECG	2
3.2.2 Subsequent ECGs	2
3.2.3 Determine if Code STEMI eligible	2
3.3 Activating RPH Code STEMI	3
3.4 ED to prepare patient for immediate transfer to RPH Cath lab	3
3.4.1 Arrange Inter-Hospital Transfer	3
3.4.2 Clinical priorities	3
3.4.3 Medications	3
3.4.4 Documentation	4
3.4.5 Escort requirements	4
3.5 Decision-making and timing considerations for reperfusion in STEMI ¹	4
4. Roles and responsibilities	4
5. Policy context	4
6. Supporting information	4
7. Compliance, monitoring and evaluation	5
8. Relevant standards	5
9. References	5
10. Glossary	5
11. Document owner	6
12. Review	6
13. Authorisation	6
Appendix 1	7
13.1 ECG findings consistent with acute coronary occlusion myocardial infarction (ACOMI) ¹	7

Before referencing this procedure document please ensure that you have the latest version from the [Policy Library](#) information hub page.

1. Purpose

Guideline for prompt diagnosis, management, and transfer of ACOMI (STEMI and STEMI-equivalent) patients suitable for urgent reperfusion. Based on the [Australian clinical guideline for diagnosing and managing acute coronary syndromes 2025](#)

2. Applicability

Adult patients diagnosed with an ACOMI in Armadale ED (either on presentation or developing while in the ED). This guideline is applicable to medical and nursing staff within the ED

3. Guidelines

3.1 Vital signs

- Vital signs including blood pressure, heart rate, respiratory rate, peripheral oxygen saturation should be recorded on arrival

3.2 ECG

3.2.1 Initial ECG

- A patient with acute chest pain or other symptoms suggestive of an acute coronary syndrome should receive a 12-lead ECG, and have the results analysed by a senior ED doctor for evidence of ACOMI, within 10 minutes of presentation
- If presenting via SJA with concerns for STEMI to remain on SJA stretcher for ECG
- Continuous cardiac monitoring is recommended while assessment for ACOMI continues in ongoing ischaemic symptoms, haemodynamic compromise or new ischaemic changes on ECG. Ensure a defibrillator is readily available

3.2.2 Subsequent ECGs

- In people who are symptomatic and/or haemodynamically compromised, ECGs should be performed at (a minimum of) 15-minute intervals until the symptoms have resolved
- For those with ongoing ischaemic symptoms and an inconclusive standard 12-lead ECG – perform right sided and/or posterior leads
- Additional ECGs should be performed if the person's symptoms reoccur, there are changes in character or a change in their clinical condition.

3.2.3 Determine if Code STEMI eligible

12-lead ECG demonstrates STEMI (confirmed by senior doctor):

ST segment elevation $\geq 1\text{mm}$ in ≥ 2 contiguous leads

AND

chest pain < 12 hours

Discuss with RPH on-call CCU registrar/consultant if:

- [STEMI-equivalent](#) pattern on ECG
- New (L)BBB

- > 85 years old
- Pulmonary oedema/shock
- Possible posterior MI
- Major co-morbidity
- Major bleeding concerns
- Out of Hospital Cardiac Arrest (OOHCA)

3.3 Activating RPH Code STEMI

- Immediate call by **senior doctor** to RPH Switchboard – 9224 2244 or *4323
- Advise “**Code STEMI**” call from Armadale for CCU registrar
- Switchboard to stay on-line to transfer to CCU registrar
- If Switchboard unable to contact CCU registrar to transfer call to CCU Consultant
- **RPH CCU registrar/consultant will activate Code STEMI via 55 if eligible**

3.4 ED to prepare patient for immediate transfer to RPH Cath lab

3.4.1 Arrange Inter-Hospital Transfer

- Manage on SJA stretcher if arrived by ambulance
- Coordinator to request a “Code STEMI” Priority 1 transfer via SJA to RPH Cath lab
- Inter-Hospital Transfer bed request via EBM for CCU at RPH
- Inform RPH ED EPIC if patient needing to go via ED rather than direct to Cath Lab

3.4.2 Clinical priorities

- Recruit a second doctor for assistance
- Avoid routine oxygen use in patients with saturations $\geq 90\%$
 - If oxygen therapy required, SpO₂ should not exceed 96%
- Continuous cardiac monitoring/defib pads attached
 - Transfer to SJA equipment when they arrive
- IV access inserted and secured (1x essential, 2x preferred)
- Ensure appropriate [medication](#) prescribed and administered
- If time allows, perform a targeted history and physical examination
- Acquiring a chest X-ray should not delay transfer for urgent reperfusion

3.4.3 Medications

- Aspirin 300mg PO (if not already given by SJA)
- Ticagrelor 180mg PO (unless thrombolysis given)
- Heparin 5000 units IV (dose reduction if < 70kg)
- GTN sublingual spray for ongoing pain if no contraindications
 - Can give every 5 minutes for up to 3 doses
 - Consider [GTN infusion](#) if ongoing pain

- Contraindications: Hypotension, Inferior STEMI, RV STEMI, use of PDE-5-Inhibitors in previous 24 hours (e.g. Viagra)
- Titrated IV Fentanyl/Morphine for analgesia

3.4.4 Documentation

- **ED medical notes to be copied into DMR**
- **ECGs**, nursing notes, medication chart etc to be photocopied prior to transfer
- Whenever possible ensure patient's next of kin is made aware of transfer to another site
- Strong consideration for discussing/documenting [Goals Of Patient Care](#)/Ceiling of Care

3.4.5 Escort requirements

- Refer to [ACEM/CICM/ANZCA Joint Guideline for the Transport of Critically Ill Patients](#) and [EMHS Inter-Hospital Transport Policy](#)
- Discuss with senior – most patients can be transferred with Paramedic Crew only
- Escort to check Adult Transfer Bag prior to leaving

3.5 Decision-making and timing considerations for reperfusion in STEMI¹

- PCI is the preferred reperfusion strategy in people with STEMI whose symptoms have lasted less than 12 hours. PCI should be performed within 120 minutes of first medical contact.
- Fibrinolysis should be performed in people with STEMI whose symptoms have lasted less than 12 hours if primary PCI cannot be delivered within 120 minutes of first medical contact. Refer to [AKG Thrombolysis Guideline](#)

4. Roles and responsibilities

- 9.1 Appendix A: IHPT Roles, Responsibilities and Escalation of [EMHS Inter-Hospital Transport Policy](#)

5. Policy context

The following documents are required to give effect to this:

- EMHS Policy
 - [EMHS Inter-Hospital Transport Policy](#)

6. Supporting information

The following documents support the implementation of this policy:

- [RPH/SJA STEMI Activation Pathway](#)
- [AKG Thrombolysis Guideline](#)
- [Australian clinical guideline for diagnosing and managing acute coronary syndromes 2025](#)

7. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated with clinical incidents being reported on Datix CIMS.

Annual auditing of:

- Immediate Management
- Early Assessment
- Timely Perfusion
- Risk Stratification

In compliance with the Acute Coronary Care Standard using the Australian Commission on Safety and Quality.

Monthly benchmarked STEMI data provided to Head of Emergency Department AKG from East Metropolitan Health Service on door to balloon, timely perfusion and identifying delays.

Reported at the Emergency Management Departmental Safety and Quality meeting.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 – Action 1.5.2 Actions are taken to minimise risks to patient safety and quality of care
- Standard 1 – Action 1.8.2 Early action to reduce the risks for at-risk patients.

Australian Commission on Safety and Quality in Health Care (ACSQHC)

- Acute Coronary Syndromes Clinical Care Standard (2019)
 - Quality Statement 1 – Immediate management
 - Quality Statement 2 – Early assessment
 - Quality Statement 2 – Timely reperfusion

9. References

1. National Heart Foundation of Australia & Cardiac Society of Australia and New Zealand: Australian clinical guideline for diagnosing and managing acute coronary syndromes 2025 Available from: <https://www.heartfoundation.org.au/for-professionals/acs-guideline?tab=0> (accessed Jan 2026)
2. Joint Guideline for Transfer of Critically Ill Patients (G03 Australasian College for Emergency Medicine, PG52 Australian and New Zealand College of Anaesthetists, IC-10 College of Intensive Care Medicine) August 2025. Available from: https://acem.org.au/getmedia/0daba691-5e60-4a88-b6a8-24f2af3e5ebf/Guidelines_for_the_Transport_of_Critically_Ill_Patients (accessed Jan 2026)

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
STEMI	ST-elevation myocardial infarction

Term	Definition
ACOMI	Acute Coronary Occlusion Myocardial Infarction
PCI	Percutaneous Coronary Intervention
OOHCA	Out Of Hospital Cardiac Arrest

11. Document owner

Director Clinical Services

Enquiries relating to this guideline may be directed to:

Title: Dr Emma Trace
 Department: Emergency Department
 Email: emma.trace@health.wa.gov.au

12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	13 03 2026	13 03 2029	Full review, significant changes throughout document.

13. Authorisation

This guideline has been authorised and issued by:

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	11 03 2026
Published date	13 03 2026

Appendix 1

13.1 ECG findings consistent with acute coronary occlusion myocardial infarction (ACOMI)¹

	Criteria	Supporting information and illustration
A. Regional STE with reciprocal STD	STE ≥ 1 mm at the J-point in two contiguous leads in all leads other than V2-4. V2-4 STE criteria: ≥ 1.5 mm in women ≥ 2 mm in men ≥ 40 years ≥ 2.5 mm in men < 40 years	
B. High lateral MI	STE I, aVL, V2 STD III (+/- II, aVF) Subtle STE V5, V6 and reciprocal changes in aVF may be seen.	
C. Posterior MI	Precordial STD ≥ 0.5 mm V1-3 Confirm with posterior leads (V7,8,9) with findings of STE: ≥ 0.5 mm in women and men ≥ 40 years ≥ 1 mm in men < 40 years	V7, 8, 9 supplementary lead placement
D. Right ventricular MI	STE ≥ 0.5 mm in any right-sided chest lead (V3R-V6R), but particularly V4R. STE ≥ 1 mm in men < 30 years	Right precordial supplementary lead placement
E. De Winter T waves	J-point depression with up-sloping ST segments and tall, prominent, symmetric T waves in precordial leads, with STE (≥ 0.5 mm) in aVR and an absence of STE in precordial leads.	
F. Modified Sgarbossa criteria (LBBB or paced rhythm)	Any of the following: A) Concordant STE > 1 mm in leads with positive QRS complex B) Concordant STD ≥ 1 mm V1-3 C) STE ≥ 1 mm in one or more leads at the J-point which is proportionally discordant to the preceding S wave by $> 25\%$.	